

(∞, n) -categories and monoidal structure on higher categories ★

Functorial QFT, Cobordism hypothesis ★

Symmetries of QFT $\mathbb{Z}$, $\text{sym}(Z) \simeq$ Topological defects using bi-coloured cobordisms. ★

Defects as supernatural transformations. ★

TQFT $\text{sym}(Z) \simeq \text{End}_{\text{direct}}(T(*))$

Higher categories

Categories enriched in categories

data: set of objects
for any two objects we have a Hom category $\text{Hom}(A, B)$
for every object A we have an object in $\text{Hom}(A, A)$
 id_A

for any triple of objects A, B, C

we have a Functor

$\circ: \text{Hom}(A, B) \times \text{Hom}(B, C) \rightarrow \text{Hom}(A, C)$

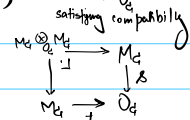
called composition
satisfying associativity and identity
axioms strictly.

strict 2-category

Categories internal to categories

we have a category $\mathcal{C}$
define a category (object inside $\mathcal{C}$).

$\mathcal{C} = (O_{\mathcal{C}}, M_{\mathcal{C}}, i: O_{\mathcal{C}} \rightarrow M_{\mathcal{C}}, s, t: M_{\mathcal{C}} \rightarrow O_{\mathcal{C}}, \circ: M_{\mathcal{C}} \times_{O_{\mathcal{C}}} M_{\mathcal{C}} \rightarrow M_{\mathcal{C}})$
satisfying compatibility.



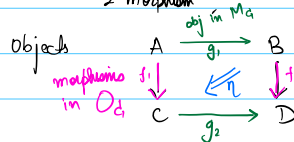
What is a category object in Set? A kitten.

What is a category object in Cat? A double category

2 notions of 1-morphism

morphism in $O_{\mathcal{C}}$ and objects in $M_{\mathcal{C}}$.

2-morphism

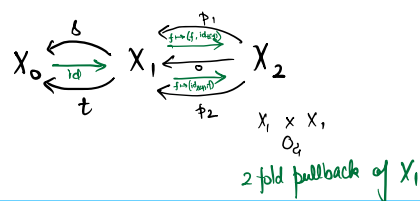


η is a morphism in $M_{\mathcal{C}}$ from $f_1 \rightarrow f_2$
such that $s(\eta) = f_1$
 $t(\eta) = f_2$.

invertible obj's in $M_{\mathcal{C}}$ (with composition) are identified with morphisms in $O_{\mathcal{C}}$

What is a category object in the category of groupoids.

2 cells (η) and morphisms in $O_{\mathcal{C}}$ become invertible. To obtain a (2,1)-category, we impose a completeness condition.



Δ

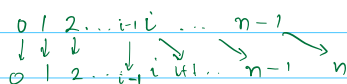
1-category obj : finite totally ordered sets (non-empty)
 mor : order preserving maps.

Note: every object is isomorphic to an object of the form $[n]$ $0 \leq i \leq 2 \dots \leq n$.

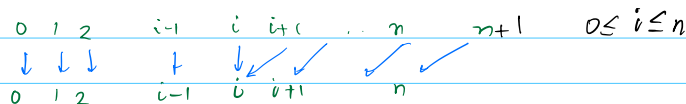
A simplicial object in a category: a functor from $\Delta^{op} \rightarrow \mathcal{C}$
 A simplicial set is a simplicial object in Set .

Th: Δ is generated by the following morphisms.

$$\delta_i^n : [n-1] \rightarrow [n] \quad 0 \leq i \leq n-1$$



$$\sigma_i^n : [n+1] \rightarrow [n]$$



Relations.

$$\delta_i^{n+1} \circ \delta_j^n = \delta_{j+1}^{n+1} \circ \delta_i^n \quad i \leq j$$

$$\sigma_j^n \circ \sigma_i^{n+1} = \sigma_i^n \circ \sigma_{j+1}^{n+1} \quad i \leq j$$

$$\sigma_j^n \circ \delta_i^{n+1} = \begin{cases} \delta_i^n \circ \sigma_{j-1}^{n-1} & i < j \\ \text{id}_{[n]} & i = j \text{ or } i = j+1 \\ \delta_{i-1}^n \circ \sigma_j^{n-1} & j+1 < i \end{cases}$$

Let X be a simplicial set. Then $X([n]) = X_n$

set of n -simplices in X .

$$X(\delta_i^n) = d_i^n : X_n \rightarrow X_{n-1}$$

$$X(\sigma_i^n) = s_i^n : X_n \rightarrow X_{n+1}$$

the 'face' map
 the 'degeneracy'

Space:

Starting point: An $(\infty, 1)$ -category of ∞ -groupoids

1. construction in higher topos theory.

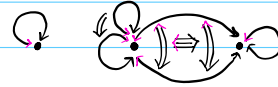
obj: CW complexes

1-mor: continuous maps.

2-mor: homotopies

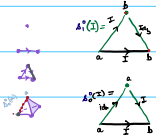
...

'Think of points in a space, paths between them, homotopies between paths...' topological space CW-complex



2. A simplicial object in a category $\mathcal{C}$

is a collection of objects X_n for $n \in \mathbb{N}$ and morphisms.



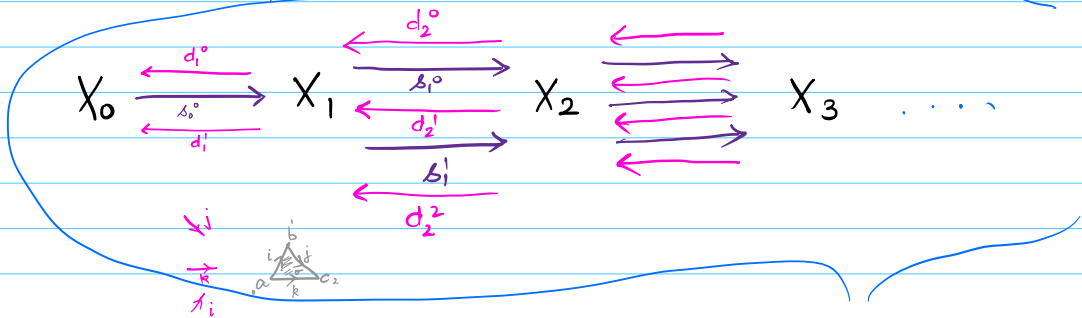
$$d_n^i: X_n \rightarrow X_{n-1} \quad 0 \leq i \leq n, n > 0$$

$$s_n^i: X_n \rightarrow X_{n+1} \quad 0 \leq i \leq n, n \geq 0$$

such that

Think:

"points" of X_n should be thought of as n -simplices
 d_n^i joins face opposite i -th vertex
 s_n^i takes an n -simplex and thinks of it as a degenerate $n+1$ -simplex, there are $n+1$ ways to do this



Let us look at simplicial object in Space X

an $(\infty, 1)$ -category

X_0 : ∞ -groupoid of objects of X

X_1 : ∞ -groupoid of 1-morphisms (length 1-chain) in X

X_n : ∞ -groupoid of composable (length n -chains) in X

} ← source of higher morphisms.

X_2 ← composition

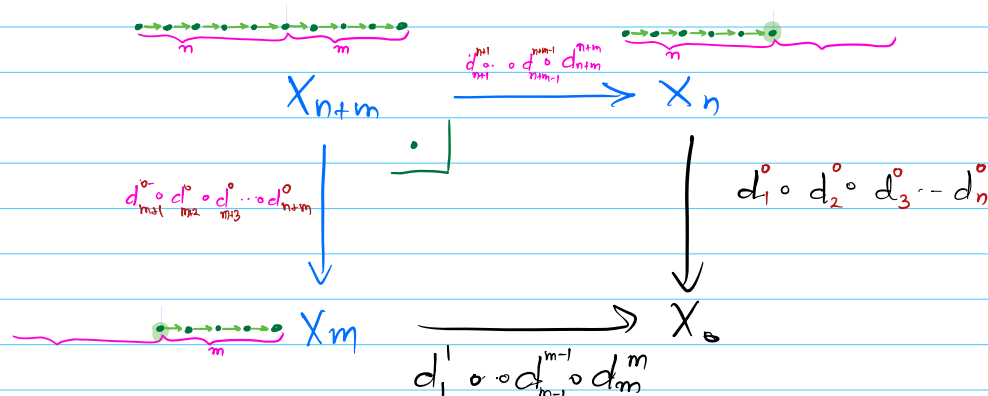
X_3 ← associativity of 3-morphisms

...

X_n should be related to X_1

The Segal Condition

choose n, m



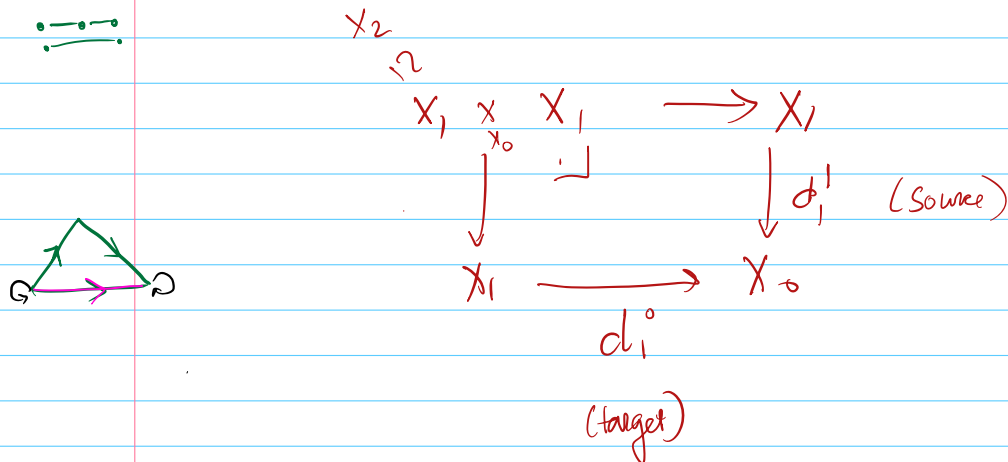
a homotopy pullback commutes up to homotopy

A simplicial object in $(\text{Space})^{\mathcal{C}}$ that satisfies the Segal condition is a Segal (space) object in $\mathcal{C}$.

Problem: 2 'types' of 1-morphisms.

1-morphisms of X_0 and 'points' of X_1
(ω -groupoid)

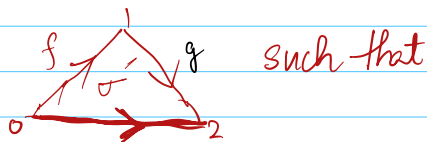
Composition : $X_2 \longrightarrow X_1$
 d_2^1



$$X_0 \xrightarrow{\delta_0^0} X_1$$

identity -

let f be an object of X_1 , we say f has a left inverse if there is a σ in X_2



$$s_0^\circ(a)$$

$a \xrightarrow{id} a$
 $S^1(j)$

A hand-drawn triangle with vertices labeled 'a', 'b', and 'c'. Vertex 'a' is at the bottom left, 'b' is at the top, and 'c' is at the bottom right. The triangle is drawn with simple black lines on a white background.

$$d_2^2(\sigma) \stackrel{?}{=} f$$

1-morphism in X

$$d_2'(\sigma) \simeq \mathcal{S}_0 \circ (d_1'(f))$$

velocity or source (f).

• $d_2^0(\sigma)$ is the left inverse of f .

f is invertible if it admits both left and right inverses

get a sub-groupoid of X_1 consisting of invertible 1-morphisms ('points' of X_1) $\cup \cup := X_1^{ev}$

$$X_0 \xrightarrow{\beta_0} X_1$$

Diagram illustrating the relationship between x_1 and x_2 in the context of the simplex method. The diagram shows a vector labeled x_1 and a vector labeled x_2 originating from the same point. A curved arrow indicates a rotation or transformation from x_1 towards x_2 .

complete Segal space
(object)

$(\infty, 1)$ -category

Now let us take simplicial objects inside $(\infty, 1)$ -category of complete Segal spaces.

$$\Delta^{op} \rightarrow \text{Func}(\Delta^{op}, \text{Space})$$

$\vdots$ n-times

$$\Delta^{op} \rightarrow \text{Func}(\Delta^{op}, \text{Func}(\Delta^{op}, (\dots \text{Func}(\Delta^{op}, \text{Space}))))$$

n-fold complete Segal space $(\Delta^{op})^n \rightarrow \text{Space}$

is a model for an (∞, n) -category

element $\mapsto$ morphism
monoid $\leadsto$ category
M BM

Symmetric

Monoidal structure?

★

using
philosophy
of going deeper
by taking
universal objects

A monoid object X in $\mathcal{C}$ (∞, n) -category

is a Segal object such that $X_0 \xrightarrow{\sim} 1$

$$\Delta^{op} \rightarrow \mathcal{C}$$

Segal condition

monoidal category $\leadsto$ 2-category with 1 object

terminal object of the category

Segal object in Set is a category.

$\therefore$ A monoidal (∞, n) category is a monoid object in the category of (∞, n) -category, already gives us tensor and associativity (composition) given order coherence

$\rightarrow$ ~~Symmetric~~ monoid object $X: \Delta^{op} \rightarrow \mathcal{C}$ satisfying Segal condition and X_0 contractible.

inside X

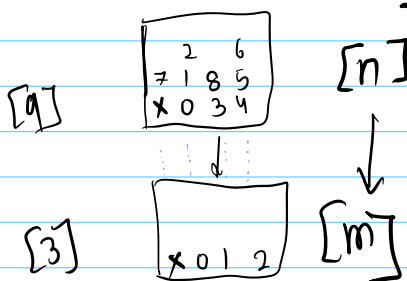
we want to be able to take finite tuples of 'points' in X_1 and 'know' them in a way that order of knowing does not matter.

Remind what Δ was 'finite sets and order preserving maps'

We now take Γ^{op} 'finite sets and maps' base point preserving

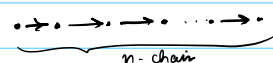
$$\begin{matrix} [n] \\ [0] & [1] & [2] & \dots & [n] \\ \{*\} & \{*, 0\} & \{*, 0, 1\} & \dots & \{*, 0, 1, \dots, n-1\} \end{matrix}$$

$$X : \Gamma^{op} \rightarrow \mathcal{C}$$

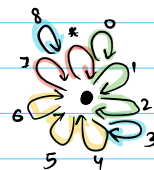


X_n

X_m



n-chain
n-tuple of 1-morphisms.
because X_0 is contractible
any two 1-mor have
same source and target



$$\begin{array}{rcl} 0 \cdot 1 \cdot 2 & 0 \otimes 1 \otimes 2 & 0 \\ \hline 3 \cdot 8 & 3 \otimes 8 & 1 \\ \hline 4 \cdot 5 \cdot 6 & 4 \otimes 5 \otimes 6 & 2 \end{array}$$

Symmetric (monoidal) object $X : \Gamma^{op} \rightarrow \mathcal{C}$

satisfying segal condition
and X_0 contractible.

Sym monoidal (∞, n) -category $C : \Gamma^{op} \rightarrow n$ -fold complete segal space

$$C : \Gamma^{op} \rightarrow \text{Func}((\Delta^{op})^n, \text{Space})$$

$$C : \Gamma^{op} \times (\Delta^{op})^n \rightarrow \text{Space}$$

QFT (∞, d) -
"smooth" $\otimes$ -functor
in-abel

fix geometric structure, fix target
usually n -Vect
 $\vdots$
 $\Omega^{n-1}(\mathbb{1}) = \text{Vect}$

$$\text{Cob}_d^g \longrightarrow \mathcal{V}$$

(∞, d) -category

want
 x



codim 1



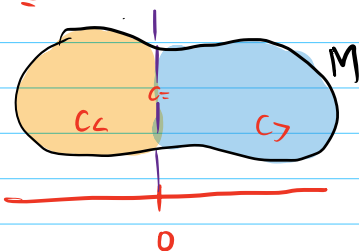
* start with M (d -dimensional)

A cut is a triple $(C_<, C_<=, C_>)$
of subset of M such that

there is $h: M \rightarrow \mathbb{R}$ smooth

regular at 0
 $dh|_p: T_p M \rightarrow T_0 \mathbb{R}$
is surjective
 $\forall p \in h^{-1}(0)$

$$C_<= = h^{-1}(0), C_< = h^{-1}(-\infty, 0), C_> = h^{-1}(0, \infty)$$



A cut $[m]$ -tuple is a collection of cuts $C_j = (C_j^<, C_j^<=, C_j^>)$
 $j \in [m]$ such that

$$C_{\leq 0} < C_{\leq 1} < \dots < C_{\leq m}$$

defm. $\text{Cob}_d^g : (\underline{\Delta^{\text{op}}})^d \times \underline{\Gamma^{\text{op}}} \rightarrow \text{Space}$

for fixed $\underline{m} \in \Delta^d$, $\langle l \rangle \in \Gamma$
 $([m_1], \dots, [m_d])$

take $m, \langle l \rangle$ and produce a space

Objects $X(\underline{m}, \langle \ell \rangle)$

$$X(\underline{m}, \langle \ell \rangle)$$

← let this have a geometric structure.

* d -dim manifold M

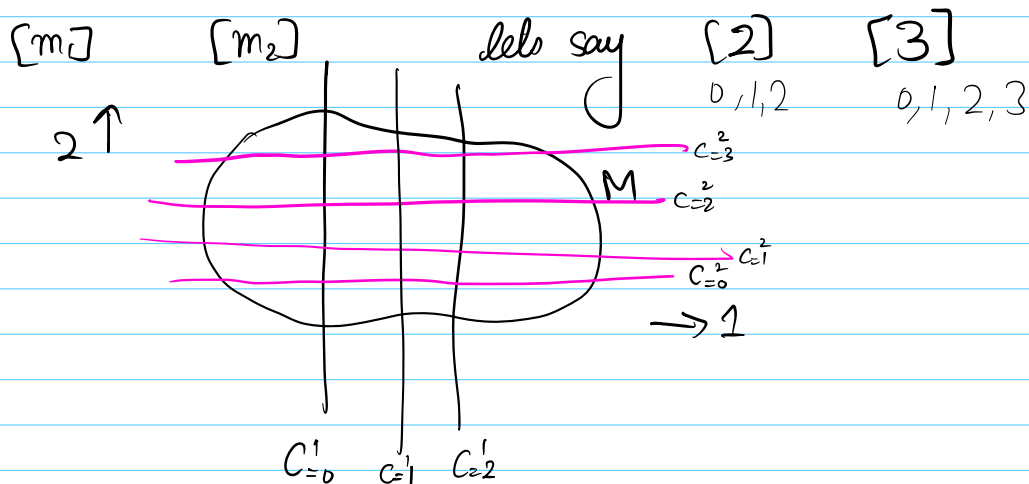
- * d -dim manifold M
- * $\forall i \leq d$, a cut $[m_i]$ tuple C^i ^{length} $[m_i]$ in i -th direction

- * A map $P: M \rightarrow \langle l \rangle$ partitioning the set of connected components into l disjoint subsets

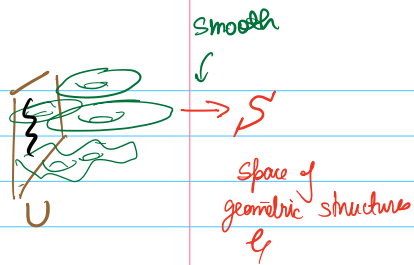
1-Merib

2-morfo

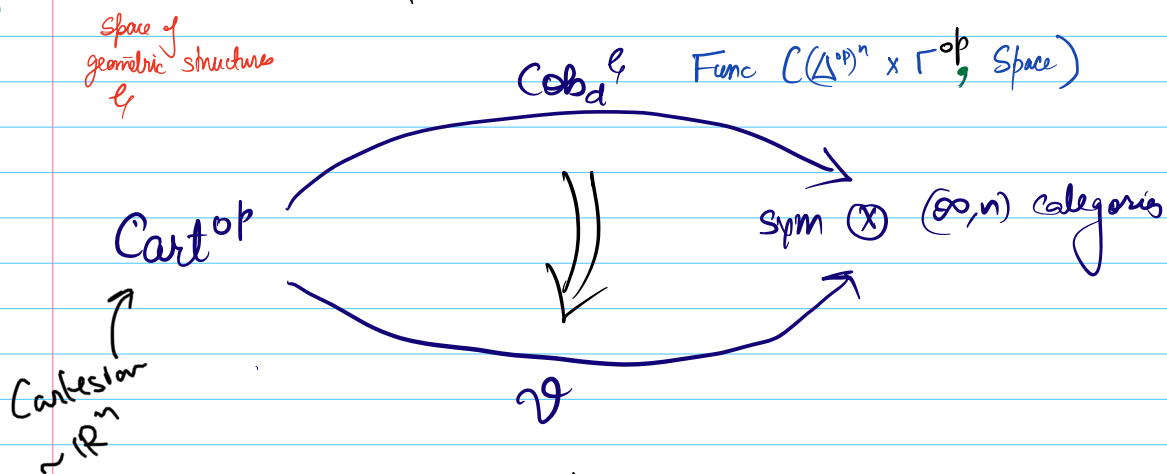
Transversality $\rightarrow$ Space of cut respecting diffeomorphisms

$$d=2$$


smoothly varying the geometric structure



$$\text{Cob}_d^{\mathcal{L}} \longrightarrow \mathcal{V}$$



$$\text{Cob}_d^{\mathcal{L}} : (\Delta^{op})^n \times \Gamma^{op} \times \text{Cart}^{op} \longrightarrow \text{Spm} \otimes (\infty, n) \text{ categories}$$

or

$$\text{PSH}(\Delta^a \times \Gamma \times \text{Cart})$$

Spaces

For TQFTs

Cobordism Hypothesis:

$$\text{Func}^{\otimes}(\text{Cob}_d^{\text{fr}}, \mathcal{V}) \xrightarrow{\text{ev}_*} \mathcal{V}$$

\mathbb{Z} \curvearrowright

\Rightarrow \text{Core}(\mathcal{V}^{\text{fr}, d})

lets unpack

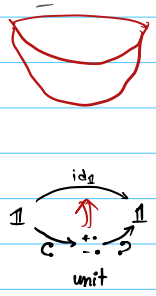
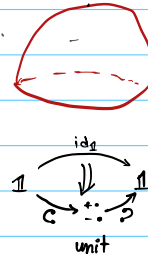
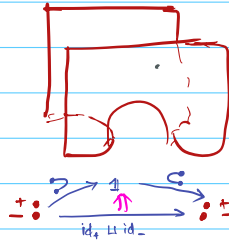
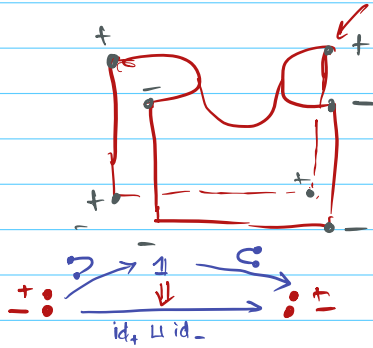
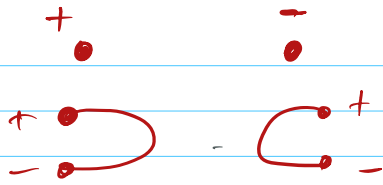
maximal
sub n-manifold

objects have duals, 1-dualizable, ev, coev exist

$$X \otimes X^{\vee} \xleftarrow{\text{coev}} 1 \quad X^{\vee} \otimes X \xrightarrow{\text{ev}} 1$$

ev and coev have adjoints 2-dualizable
both left and right
unit, counit, associators exist.

have adjoints 3-dualizable etc.



$$id_A \in A \xrightleftharpoons[g]{f} B \in id_B$$

$$\eta: 1_A \rightarrow g \circ f$$

$$\varepsilon: f \circ g \rightarrow 1_B$$

f is left adjoint to g

Frobenius
alg object



$$Z: \text{Cob}_d^g \longrightarrow \mathcal{V}$$

$d\text{-dim QFT}$

Topological defects

Freed TQFT

$$Z: \text{Cob}_d \longrightarrow \mathcal{V}$$

defect of codim $\underline{l}$ in $k\text{-dim manifold } M$

$l \in \{0, \dots, k\} \quad k \in \{0, \dots, d\}$

$$\begin{matrix} \Rightarrow \dim k-l \\ D \subset M \end{matrix}$$

takes $\mathcal{V} \subset M$ to be open tubular neighbourhood of $D \subset M$.

$\mathcal{V} \rightarrow D$ fiber bundle with fiber closed l -disk.

$$p \in D$$

$$\partial \mathcal{V}_p \text{ is } S^{l-1}$$



defn. Assumption M is closed, DCM is closed

$\mathcal{D}$ A local defect- at $p \in D$

$$\mathcal{S}_p: \mathbb{1} \longrightarrow Z(\partial \mathcal{V}_p)$$

$$\mathcal{S}_p: \mathbb{1} \longrightarrow Z(S^{l-1})$$

special case

D is a point M is d dimensional

$$l = d$$

$$\mathcal{S}_p: \mathbb{1} \longrightarrow Z(S^{d-1})$$

$$\mathcal{S}_p \in Z(S^{d-1})$$

} global
local
point
defect
in
spacetime.

2. transparent (local) defect

Ref. Topological Symmetry in Quantum Field Theory.
Freed, Moore, Teleman.

The Trinity

$$Z: \text{Cob}_d^e \rightarrow \mathcal{V}$$

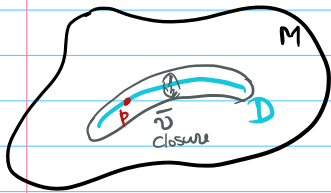
Freed 'Topological Symmetry in QFT'

take M of $\dim m \leq d$

A defect D of $\text{codim } l$ in M

D is a submanifold of M .

Assumption M is closed, D is closed sub-manifold.



$\mathcal{V}$ is the tubular neighbourhood

of full dim m .

$$\bar{\mathcal{V}} \rightarrow D$$

with fibres l dim discs.

$$p \in D$$

fibre above p is $\bar{\mathcal{V}}_p$ and $\partial \bar{\mathcal{V}}_p$ is $\simeq S^{l-1}$

1) A local defect at p $S_L \in \text{Hom}_{\Omega^{l-1}\mathcal{V}}(\mathbb{1}, Z(\partial \bar{\mathcal{V}}_p))$

2) A global defect $S_D \in \text{Hom}_{\Omega^{m-1}\mathcal{V}}(\mathbb{1}, Z(\partial \bar{\mathcal{V}}))$

3) Transparent defects.

local

$$\text{view } \bar{\mathcal{V}}_p: \phi \rightarrow \partial \bar{\mathcal{V}}_p$$

$$Z(\bar{\mathcal{V}}_p): \mathbb{1} \rightarrow Z(\partial \bar{\mathcal{V}}_p)$$

global

view

$$\bar{\mathcal{V}}: \phi \rightarrow \partial \bar{\mathcal{V}}$$

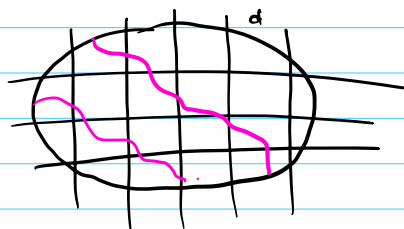
$$Z(\bar{\mathcal{V}}): \mathbb{1} \rightarrow Z(\partial \bar{\mathcal{V}})$$

define as evaluation of a global defect

$$\mathbb{1} \xrightarrow[\mathcal{S}_D]{Z(M)} Z(\partial \bar{D}) \xrightarrow[Z(M \setminus \bar{D})]{} \mathbb{1}$$

2. $\text{Sym}(Z)$ is a (or $d-1$) sym monoidal category

obj $D: \text{Cob}_d^g, \text{codim } 1 \text{ defects} \rightarrow \mathcal{V}$



when restricted
give Z

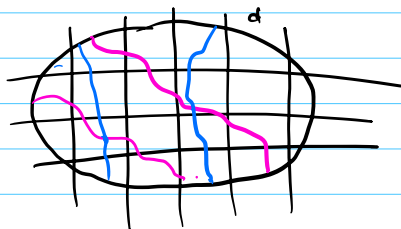
1-mor

Sym
monoidal
functors

$\text{Cob}_d^g, \text{codim } 2 \text{ defects between codim } 1 \text{ defects}$

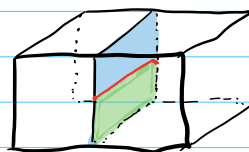
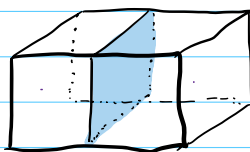
$\rightarrow \mathcal{V}$

$\text{Hom}(D_1, D_2)$



when restricted
gives blue and green
codim 1 functors

when restricted further
give Z



2-mor

$\text{Cob}_d^g, \text{codim } 3 \text{ defects between } 2 \text{ codim } 2 \text{ defects}$

$\rightarrow \mathcal{V}$

that restrict
well

⑧

$D_1 \otimes D_2$

3. Supernatural transformation

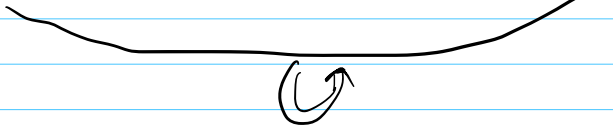
$$\text{Cob}_{d-1}^{\square} \hookrightarrow \text{Cob}_d^{\square} \xrightarrow{z} \mathcal{V}$$

codim 1 defect

$$\text{End}(\text{Cob}_{d-1}^{\square}, \mathcal{V})$$

$$\searrow \quad \nearrow$$

$$\text{Cob}_d^{\square}$$

$$\text{Cob}_k \hookrightarrow \text{Cob}_{k+1} \hookrightarrow \dots \hookrightarrow \text{Cob}_d \xrightarrow{\quad} \mathcal{V}$$


Theorem :

$$\text{Cob}_2^{\text{domain walls}} \longrightarrow \mathcal{V}$$

2D TFT with codim 1 defect
in the bulk

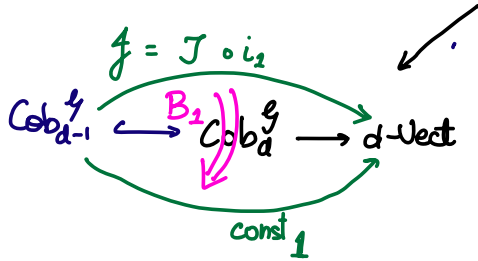
||

Natural transform. $z \circ i \Rightarrow z \circ \hat{i}$

QFT d-dimensional

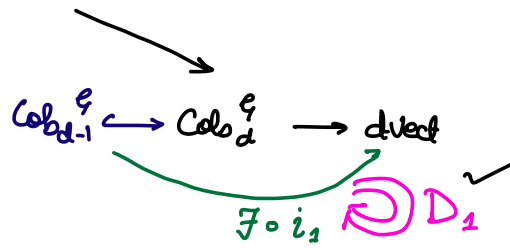
$$J : \text{Cob}_d^g \rightarrow d\text{-Vect}$$

$$\star \Omega^{d-1} d\text{-Vect} = \text{Vect}$$



on objects in Cob_{d-1}^g

$$B_1(*) \in \text{Hom}_{d\text{-Vect}}(f(*), 1)$$



d-1 $\otimes$ -category

$$D_1(*) \in \text{Hom}_{d\text{-Vect}}(f(*), g(*))$$

$$D_1(*) \in \text{End}_{d\text{-Vect}}(f(*))$$

naturality of D_1

$$\forall I \in \text{Hom}_{\text{Cob}_d^g}(*_1, *_2)$$

$$\begin{array}{ccc} f(*_1) & \xrightarrow{f(I)} & f(*_2) \\ D_1(*) \downarrow & \searrow D_1(I) & \downarrow D_1(*) \\ g(*_1) & \xrightarrow{g(I)} & g(*_2) \\ \vdots & & \vdots \end{array}$$

$$\text{End}_{d\text{-Vect}}(f(*)) \text{ same as}$$

$$\star \text{End}_{d\text{-Vect}}(J(*))$$

d-1 $\otimes$ -category.

composition of morphisms

$$f \otimes g = g \circ f \text{ for objects}$$

for TQFT

same as $\text{Sym}(Z)$

using 'general' cobordism hypothesis

Ref. On Classification of Topological Field Theories
Sec. 4.3 Manifolds with singularities.
Lurie